

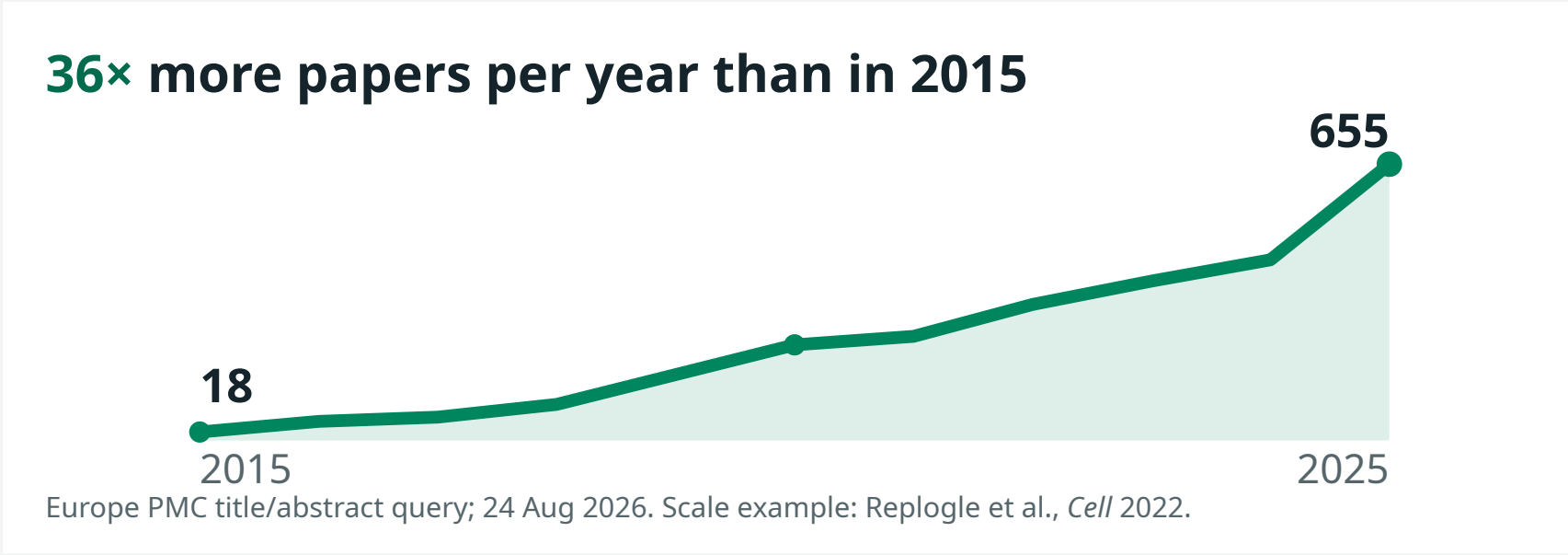
An Integrated Resource for Exploring the Impact of Genetic Perturbations

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Perturbation evidence is multiplying faster than it can be organised

One genome-scale Perturb-Seq study profiled **more than 2.5 million cells**—but reuse still begins by reconciling incompatible studies.

Repositories • identifiers • metadata depth • score semantics • processing choices



The Catalogue makes studies interoperable without detaching results from context

[illegible]

The production homepage combines target search with current coverage and biological-context summaries.

Data Modality:	Portuash.org
Perturbation Type:	CRISPRi
▼ Other Metadata Fields	
Associated Dataset:	Raw data for Juxtal Pertuash seq experiment: GSE254455 juxtal_seq_experiment_01 juxtal
Library Genes Per Target:	2
Library Lactwood Generations:	2
Library Manufacturers:	Wolfram Lab
Library Names:	control
Library Total Genes:	5344
Library Total Variants:	N/A
Library Uris:	N/A
Method UK:	N/A
Number Of Perturbed Samples:	262935
Number Of Perturbed Targets:	2391
Score Interpretation:	N/A
Timepoints:	Day 7
Cell Line:	JURKAT cell
Cell Type:	T cell
Developmental Stage:	adolescent
Disease:	Toxic childhood acute lymphocytic leukemia
Enzyme Expression Control:	constitutive transgene expression

One Catalogue already connects 1,237 datasets across three complementary modalities



1,201 CRISPR screens

gene perturbation × phenotype or fitness

26 Perturb-Seq datasets

perturbation × single-cell transcriptome

10 MAVE datasets

19,235
human targets

36
tissues

34
cell types

1,200
cell lines

Browse Targets

Explore all gene targets across perturbation experiments. Use the search bar to find genes by symbol, Ensembl ID, or name, and the facets to narrow down results.

License

☐ CC BY 4.0 (32)
 ☐ CC BY-NC-SA 4.0 (31)
 ☐ MIT License (31)
 ☐ free to use license (9)

Data Modalities

☐ CRISPR screen (32)
 ☐ Perturb-seq (31)

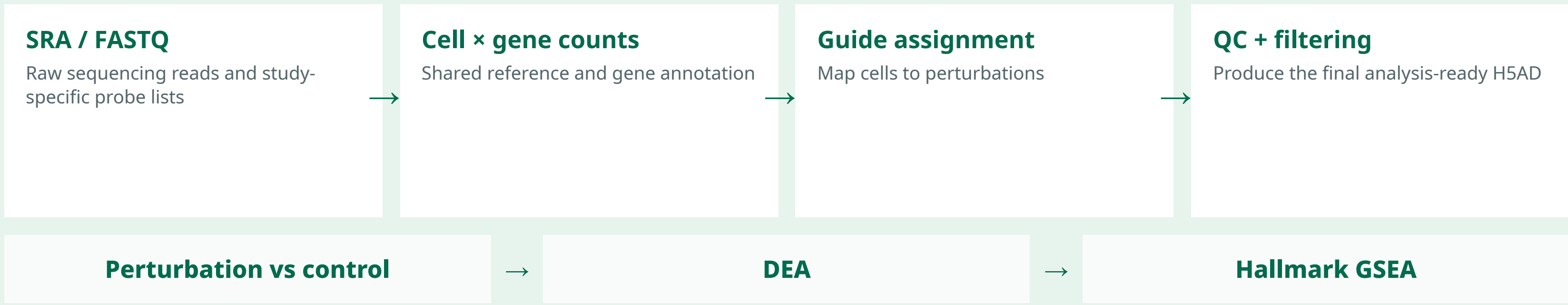
Download Metadata

TARGET NAME	PERTURB-SEQ DATASETS	CRISPR-SCREEN DATASETS	MAVE DATASETS	TOP GSEA TERMS	DATA MODALITIES
BRCA2 ENSG000000139618	Datasets: 7 (Sig. genes up ↑: 127; Sig. genes down ↓: 322)	Datasets: 1,191 Sig. hits: 640	0	HALLMARK_MYC_TARGET... HALLMARK_E2F_TARGETS HALLMARK_OXIDATIVE_P... HALLMARK_G2M_CHECKPO... HALLMARK_MITOTIC_SPL... HALLMARK_MYC_TARGET...	CRISPR screen Perturb-seq

The BRCA2 view connects contributing datasets, significant CRISPR hits, Perturb-Seq effects and enriched Hallmark pathways. Production snapshot: 24 August 2026.

Perturb-Seq reprocessing starts again from raw reads

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.



Five datasets are live. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Every result is programmatically accessible

Search, metadata, modality results and complete downloads use the same production backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /dataset/nadig_2025_jurkat
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
GET /v1/perturb-seq/nadig_2025_jurkat/download?format=csv.gz
```

```
{
  "total_rows_count": 30078802,
  "results": [
    {
      "perturbation": {
        "gene_symbol": "TFAM",
        "perturbed_target_ensg": "ENSG00000108064"
      },
      "effect": {
        "gene_symbol": "MT-ND5",
        "log2fc": -1.926,
        "padj": 0.0,
        "direction": "decreased"
      }
    }
  ]
}
```

Next, integration becomes continuous and the analytical foundation becomes stronger

Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.

Bring us the next hard perturbation-data problem

Suggest datasets, challenge the schema or develop the resource with us. We welcome scientific partnerships and funded multi-year collaborations around your programme's modalities, analyses and functionality.

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